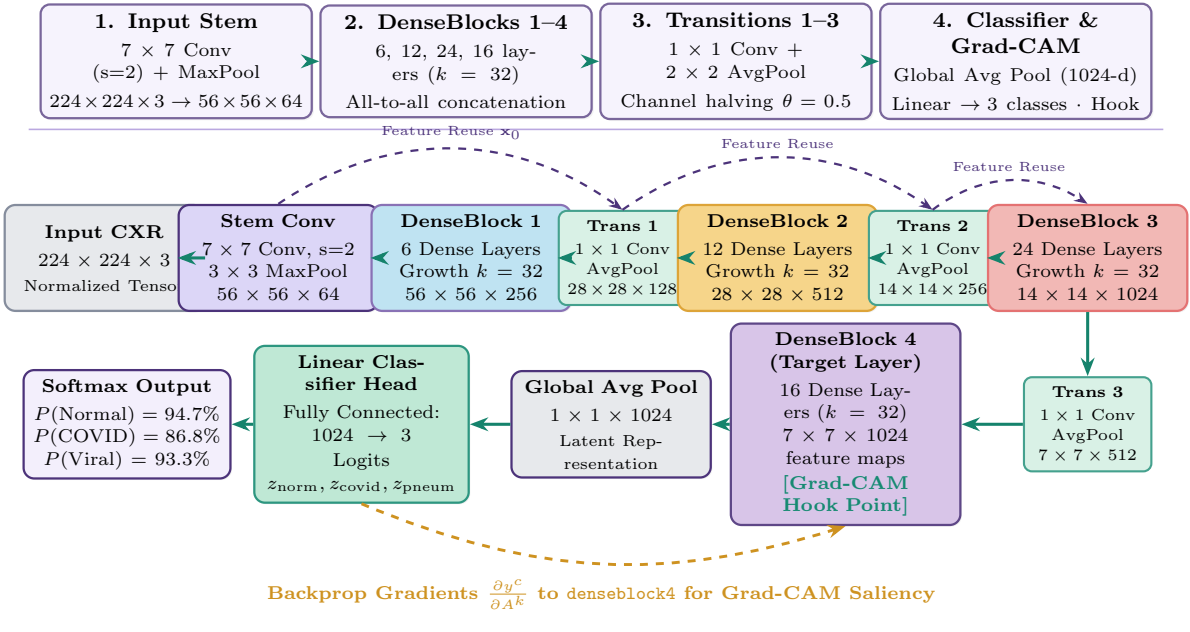

1. Input Stem
7 × 7 Conv
(s=2) + MaxPool
224 × 224 × 3 → 56 × 56 × 64
2. DenseBlocks 1–4
6, 12, 24, 16 layers (k = 32)
All-to-all concatenation
3. Transitions 1–3
1 × 1 Conv +
2 × 2 AvgPool
Channel halving θ = 0.5
4. Classifier & Grad-CAM
Global Avg Pool (1024-d)
Linear → 3 classes · Hook
Feature Reuse x0
Feature Reuse
Feature Reuse
Input CXR
224 × 224 × 3
Normalized Tenso
Stem Conv
7 × 7 Conv, s=2
3 × 3 MaxPool
56 × 56 × 64
DenseBlock 1
6 Dense Layers
Growth k = 32
56 × 56 × 256
Trans 1
1 × 1 Conv
AvgPool
28 × 28 × 128
DenseBlock 2
12 Dense Layers
Growth k = 32
28 × 28 × 512
Trans 2
1 × 1 Conv
AvgPool
14 × 14 × 256
DenseBlock 3
24 Dense Layers
Growth k = 32
14 × 14 × 1024
Trans 3
1 × 1 Conv
AvgPool
7 × 7 × 512
DenseBlock 4
(Target Layer)
16 Dense Layers (k = 32)
7 × 7 × 1024
feature maps
[Grad-CAM Hook Point]
Global Avg Pool
1 × 1 × 1024
Latent Representation
Linear Classifier Head
Fully Connected:
1024 → 3
Logits
znorm, zcovid, zpneum
Softmax Output
P(Normal) = 94.7%
P(COVID) = 86.8%
P(Viral) = 93.3%
Backprop Gradients ∂y^c/∂A^k to denseblock4 for Grad-CAM Saliency